

Minsky's Mind-Brain Musings

Minsky's writings give us a glimpse of the ideas and focus of strong-AIers over the years.

In (around) 1965: "Brains are machines made of meat," Minsky said, indicating that all we need to do, is figure out how the brain works in order to build an intelligent machine.

In 1982: Writes an essay called 'Why people think computers can't', in which he argues: "Computers can do many things that a person would have to be thinking to do. Then how could a machine seem to think but not actually think?" This illustrates the division between the strong-AIers and others. The externally visible, or the result, of any device or thing, is the only thing of importance to strong AI; others think that we need to rely upon the internal workings of something to arrive at the correct idea.

In 1985: Writes an essay, 'Communication with Alien Intelligence', in which he states:

"When first we meet those aliens in outer space, will we and they be able to converse? ... Yes, we will ... because we'll both think in similar ways." The idea here is that intelligence evolves, and that we can pass on our intelligence to our robots. Detractors would claim that machines will never be able to think, and so, there is no question of passing on our intelligence to them.

In 1991: "I think the answer lies in the fact that the brain is not merely a kind of machine, but one that is far more complex than anything ever imagined before." This reflects the growing focus in the AI community on the whole towards the understanding of the brain as a prerequisite to the building of intelligent machines.

In 1994: Writes an essay, 'Will Robots Inherit the Earth?', which he concludes with, "Will robots inherit the earth? Yes, but they will be our children. ..." The idea of augmenting our bodies with artificial tissues, organs, and so on, had already come

round, and it could very well happen that this augmentation goes to the extent where there would be no distinction between humans and robots.

In 1996, in an interview, when asked about why we have no program that has general intelligence that can play chess reasonably, Minsky said: "Only a small community has concentrated on general intelligence. ...we have collections of dumb specialists in small domains; the true majesty of general intelligence still awaits our attack." AI and cognitive science are now parallel disciplines, each feeding the other, and beginning to attack the problem of intelligence.

In 1998, in another interview, Minsky said that computers are still stupid because they don't use common-sense engines, and that not enough people are working on common-sense engines. A potentially smart machine must be fed with all the common sense we have, which we learn in the course of our lives.

recognition problems. That effectively ended funding for ANN research. Another setback came in the early 70's; the Lighthill Report convinced the British government to end support for AI research because there hadn't been sufficient applied success by then.

However, the 1970s saw various researchers doing pioneering work in several different aspects of what was, by now, recognised as an established field. As an example of how successful AI had been so far, the American Medical Association said in 1979 that Mycin—an automatic system for medical diagnosis—was as good as human experts. And then, ANNs made a big comeback in the 1980s: John Hopfield, an AI pioneer, came up with an algorithm which enabled ANNs to do what Minsky and Papert had said they couldn't.

In and after the 1980s, AI diversified into several very distinct disciplines—current fields of exploration include robotics, learning and reasoning, knowledge representation and extraction, computer vision, speech recognition, and several others. And in the 1990s, the cognitive sciences that deal with the study of how our minds work were born. Over the years, cognitive science has grown in importance because people are becoming increasingly interested in figuring out how human brains and minds operate.

Intelligence and AI

But Simon had said that by 1985, we would have machines that would be capable of doing anything a person could do! Where were those machines?

The pioneers at the time had grossly underestimated several things. They had not anticipated the complexity of brain functions that were revealed in the 1960s and the 1970s. Also, AI is tied up, by its very nature, with fields such as philosophy, biology, linguistics and so on. The pioneers had not given sufficient importance to the considerations and techniques that these fields offer.

The question now is, "Why is it so difficult to build an intelligent machine?" In order to build an intelligent machine, we first need to know what intelligence is. Words that come to mind are creativity, memory, thinking and so on. It happens, somewhat paradoxically, that the things that computers do best are things that some of the greatest human minds cannot do, and some of the things that computers cannot do as yet, are things that are easily done by any child. For example, computers can defeat grandmasters at chess, but robots cannot, as yet, easily find an object in a room, lift it and place it somewhere else!

Why is this? There are several reasons. The first, is that we think in ways that are

not easily definable; and in order to program a machine, everything needs to be defined. The second reason is that we filter out unnecessary inputs from the environment very easily. Therefore, we seem to easily figure out what is important and what is not—unlike a machine. For example, presented with a supposed fact that a white elephant is driving, we realise that the fact that the elephant is driving is more striking than that it is white.

Third, we seem to have an innate flair for pattern matching. For example, in the position below, it is obvious that white can draw, simply by moving the



Most AI programs use position values to play chess—and fail to see obvious patterns

